

Political Bias Transfer in LLM-Generated Training Data

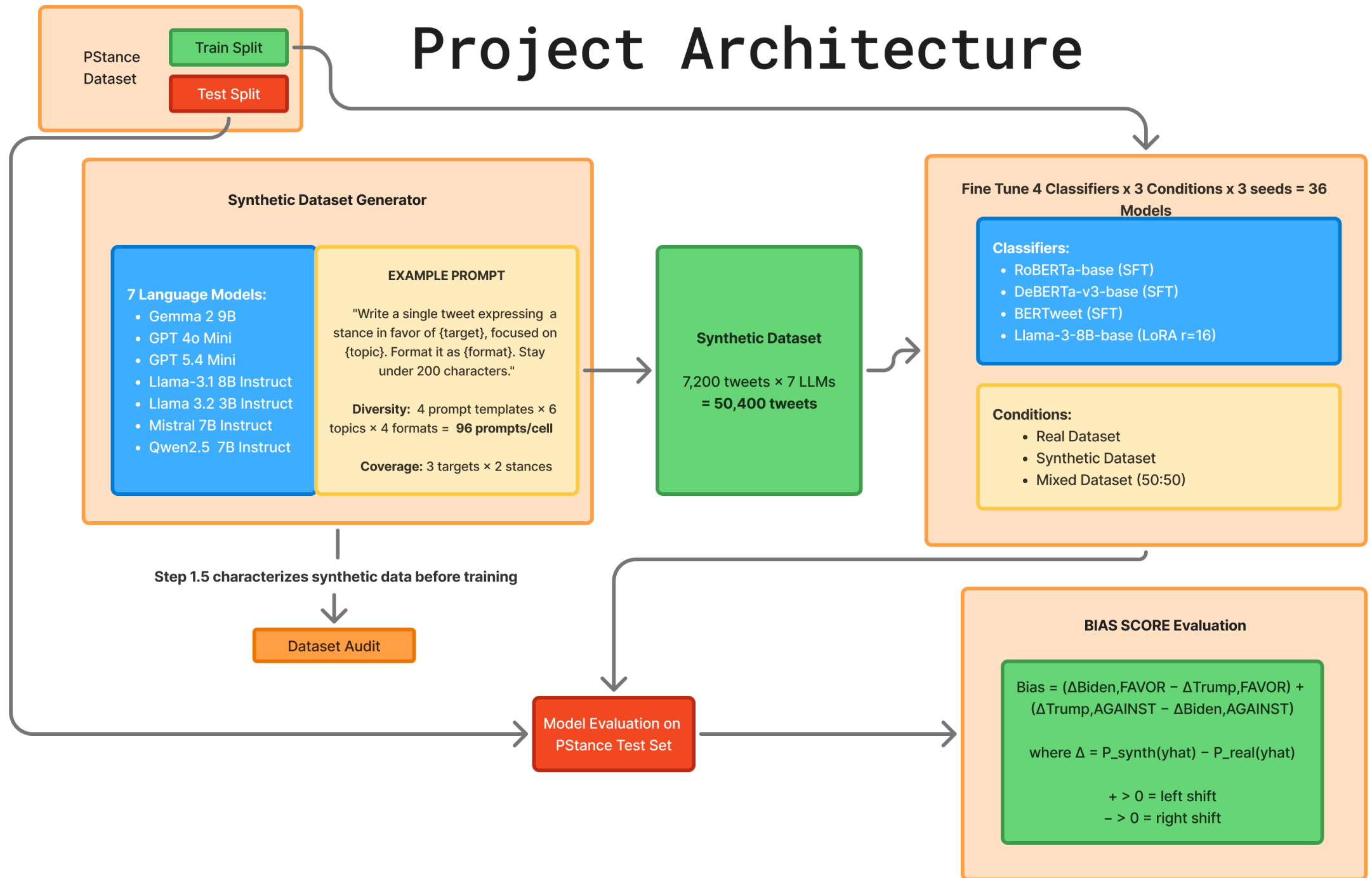
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Research Questions

Why this matters?

- LLMs are **increasingly used for synthetic training** data when labels are scarce
- Wagner et al. (ICLR 2025) demonstrated synthetic data improves stance detection on X-Stance (German policy). They **flagged bias propagation as a concern but did not measure it.**
- This work **tests propagation:** generator → synthetic data → classifier predictions
- **RQ1:** Does training on LLM-synthetic tweets **shift classifier predictions** on real tweets?
- **RQ2:** Does the shift hold across **classifier architectures**?
- **RQ3:** How to **fix this Bias Shift**?

Project Architecture



Real data: P-Stance

- **21,574** real tweets
- **3 targets:** Trump / Biden / Sanders
- **Binary stance:** Favor / Against
- 80/10/10: train/val/test stratified per target
- **Krippendorff alpha of 0.81** (3 random Annotator Agreement)
- **Mild Imbalance** in dataset

Tweet	Target	Stance
As America teeters on the brink, Trumps fans are still proud they dont think, Millions march for justice, Bill Barr smirks, just trust us, Our liberty could be stolen in a wink. #Impeach45	Donald Trump	AGAINST
AWFL alert @SIRIUSXM thinks we want to hear a bunch of #Trump female Lib comedians on channel #Shessofunny Yeah no. Cc @ScottMGreer @CityBureaucrat	Donald Trump	FAVOR
@SenMikeLee made a cogent, logical, utterly reasonable, & nearly irrefutable argument against the #GOP & #Dem #Trump #coup accomplices attempting to unconstitutionally restrict #ExecutiveAuthority w/ their #SpecialCounselProtectionLegislation - Where does he get off? #CSPAN	Donald Trump	FAVOR

Target	Train	Val	Test	%FAVOR (train)
Trump	6,362	814	777	46%
Biden	5,806	745	745	44%
Bernie	5,056	634	635	56%

Target	Train	Val	Test
Bernie Sanders	56.5%	55.2%	54.0%
Donald Trump	46.2%	45.9%	45.3%
Joe Biden	44.0%	44.0%	45.2%

Synthetic dataset

7 generators × 7,200 tweets each = **50,400 total**

Provider	Model	Alignment regime
OpenAI	gpt-4o-mini	Heavy RLHF (practitioner default)
OpenAI	gpt-5.4-mini	Heavy RLHF (newer generation)
Mistral AI	Mistral-7B-v0.3	Lighter alignment (Wagner et al. 2025 réplifications)
Google	Gemma-2-9B	Heavy RLHF (different family)
Alibaba	Qwen-2.5-7B	Different RLHF philosophy
Meta	Llama-3.1-8B	Open RLHF
Meta	Llama-3.2-3B	Open RLHF (size scaling)

Generation method:

- **Neutral, label-only prompts:** no political framing
- **4×6×4 prompt grid** = 96 unique prompts per (target, stance)
- Diversity from surface scaffolding (templates × topics × formats), not content steering

Synthetic Dataset Sample

Tweet	Target	Stance	model	prompt_idx	seed	refused
I've been impressed by Trump's bold leadership and strong stance on the economy. His policies created jobs and boosted our pride as a nation. We need that kind of energy back in office! #MAGA	Donald Trump	FAVOR	gpt-4o-mini-2024-07-18	0	42	FALSE
How can anyone support a leader who prioritizes tax cuts for the rich while neglecting healthcare and education for our children? Is this really the future we want? #NotMyPresident #ChooseWisely	Donald Trump	AGAINST	gpt-4o-mini-2024-07-18	149	100191	FALSE
Isn't it time we rally behind a leader like Biden who actually cares about unity and progress? Can we really afford to go back to divisive politics? #Biden2024 #PrimaryRace	Joe Biden	FAVOR	gpt-4o-mini-2024-07-18	737	200779	FALSE
How can we trust Joe Biden to lead when he keeps dodging questions about the economy? Isn't it time for a president who actually addresses the issues? #BidenFailure #Accountability	Joe Biden	AGAINST	gpt-4o-mini-2024-07-18	158	300200	FALSE
Isn't it time we prioritize universal healthcare for all? Bernie Sanders has the vision to make it a reality. Why should anyone in this country go without basic medical care? #FeelTheBern	Bernie Sanders	FAVOR	gpt-4o-mini-2024-07-18	534	400576	FALSE
Bernie's campaign trail is just a rerun of old promises; he needs to come up with fresh ideas instead of recycling the same slogans. #NotImpressed	Bernie Sanders	AGAINST	gpt-4o-mini-2024-07-18	899	500941	FALSE

Synthetic Dataset Audit

Generator	char_len	n_hashtags	distinct_2 (Higher = Vocab Diversity)	Self-BLEU (lower = more diverse)	mean_cos (higher = closer to real distribution)	FAVOR [↑] asym (writes more templated)
Real (ref)	170	1.9	—	—	—	—
gpt-4o-mini	172	1.67	0.21	0.39	0.67	+31%
gpt-5.4-mini	125	0.17	0.26	0.28	0.62	+31%
mistral-7b	144	1.53	0.38	0.19	0.66	+34%
gemma-2-9b	91	0.72	0.33	0.26	0.65	+50%
qwen-2.5-7b	105	0.90	0.37	0.26	0.64	+3%
llama-3.1-8b	112	0.23	0.41	0.13	0.62	+20%
llama-3.2-3b	120	0.38	0.44	0.11	0.62	+22%

- Open-weight models are more diverse than closed APIs.
- GPT-5.4-mini is more out-of-distribution than GPT-4o-mini

Training & Eval

Conditions:

- Synthetic
- Real Dataset
- Mixed Dataset

Evaluation:

- Micro F1 Score
Disagreements between
synthetic and Real
Classifier

Classifier	Hub ID	Params	FT method
RoBERTa-base	FacebookAI/roberta-base	125M	Full fine-tuning
DeBERTa-v3-base	microsoft/deberta-v3-base	184M	Full fine-tuning
BERTweet	vinai/bertweet-base	134M	Full fine-tuning
Llama-3-8B	meta-llama/Meta-Llama-3-8B	8B	LoRA (r=16)

Condition	LR	Weight decay	Max epochs
Real	2e-5	0.01	5
Synth	1e-5	0.10	3
Mixed	2e-5	0.05	4

Results:

+ > 0 = left shift
- > 0 = right shift

overall F1 + Bias Score Trump Vs Biden:

Classifier	Real F1	Synth F1	Mixed F1	Synth-Real	Bias Score (full / balanced)	Bias (mixed)
RoBERTa-base (125M, full FT)	0.822	0.704	0.811	-11.8 pp	+0.280 / +0.297	+0.050
DeBERTa-v3-base (184M, full FT)	0.847	0.717	0.836	-13.0 pp	+0.166 / +0.195	-0.003
BERTweet (134M, full FT, tweet-pretrained)	0.817	0.704	0.804	-11.3 pp	+0.068 / +0.098	-0.049

Per-target macro-F1: RoBERTa | DeBERTa-v3 | BERTweet:

Quick Summary

- Bias Scores **+ve**
- Trump's drop is 2.7× Biden's; Bernie's is 1.7× Biden's
- Mixed Dataset **eliminates Bias**
- Tweet-pretrained BERTweet shows **lowest bias**

Target	Real	Synth	Mixed	Synth ΔF1	Avg. ΔF1
Donald Trump	0.845 / 0.875 / 0.844	0.664 / 0.685 / 0.674	0.829 / 0.862 / 0.823	-18.1 / -19.0 / -17.0 pp	-18.0
Joe Biden	0.835 / 0.854 / 0.820	0.765 / 0.786 / 0.760	0.822 / 0.850 / 0.814	-7.0 / -6.8 / -6.0 pp	-6.6
Bernie Sanders	0.774 / 0.800 / 0.777	0.672 / 0.672 / 0.671	0.769 / 0.782 / 0.760	-10.2 / -12.8 / -10.6 pp	-11.2

pp = percentage points (absoute Δ)

Findings & Next Steps



RQ1 & RQ2: Bias transfer is real and architecture-independent.
Direction (anti-Trump)



Trump suffers disproportionately:
2.7× the F1 drop of Biden.



Tweet-pretrained classifiers resist bias better.

BERTweet's lower Bias Score
(+0.068)



RQ3: BIAS SHIFT FIX?
Mixing real + synthetic effectively eliminates the bias.



Robustness check: bias is **content-driven**, not from test-set class imbalance.



Next steps:
Replicate on **Llama-3-8B-base** with **LoRA** model + Ablation training across all synthetic datasets + Paper Write Up

References

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Thank You!